BUSITEMA UNIVERSITY AI/NLP INTERNISH PROGRAM

DAY 3: SENTIMENT ANALYSIS REPORT

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1. Introduction

Sentiment analysis is a Natural Language Processing (NLP) task that involves determining whether a piece of text expresses a positive or negative opinion. It is widely used in applications such as movie review analysis, customer feedback evaluation, and social media monitoring.

This project focuses on building and comparing classical machine learning models for sentiment classification using TF-IDF feature extraction.

2. Dataset

The dataset used in this project consists of labeled text data for sentiment classification.

- Total samples: 25,000
- Features:
  - Text: Movie review content
  - Sentiment: Binary label (0 = Negative, 1 = Positive)

Preprocessing Steps:

- Renamed columns for clarity ("text", "sentiment")
- Removed missing values
- Split dataset into training (80%) and testing (20%) sets
- Converted text into numerical features using TF-IDF

3. Methodology

3.1 Feature Extraction

TF-IDF (Term Frequency–Inverse Document Frequency) was used to transform text into numerical vectors. This technique assigns higher importance to meaningful words while reducing the impact of commonly occurring words.

3.2 Models Used

The following classical machine learning models were implemented:

1. Logistic Regression
2. Naive Bayes (MultinomialNB)
3. Support Vector Machine (SVM)
4. Random Forest

All models were trained using the same training data and evaluated on the same test set to ensure fair comparison.

4. Evaluation Metrics

The models were evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1-score

5. Results

Model	**Accuracy**
Logistic Regression	0.8856
Naive Bayes	0.8596
SVM	0.8844
Random Forest	0.8378

6. Discussion

The results show that Logistic Regression achieved the highest accuracy (0.8856), closely followed by SVM (0.8844). This indicates that linear models perform very well for text classification tasks, especially when using TF-IDF features, which produce high-dimensional sparse data.

Naive Bayes also performed well (0.8596), demonstrating its effectiveness in text classification despite its simplifying assumption of feature independence.

Random Forest achieved the lowest accuracy (0.8378) among the models. This is expected because tree-based models generally struggle with sparse high-dimensional data compared to linear models.

Overall, the results confirm that simpler linear models are highly effective for sentiment analysis tasks using classical approaches.

7. Conclusion

This project demonstrated the effectiveness of classical machine learning models for sentiment analysis. Among the models tested, Logistic Regression performed best, followed closely by SVM.

These results suggest that linear models combined with TF-IDF features are strong baselines for text classification tasks. Future improvements could include hyperparameter tuning, incorporating n-grams, or experimenting with deep learning models such as BERT.